

UNIVERSITY OF KARACHI

Department: UBIT



NeuroFlow CFD Analyst

A Computational Fluid Dynamics-Based Hemodynamic
Analysis Dashboard for Cerebral Aneurysm Rupture-Risk
Assessment

FINAL YEAR PROJECT DOCUMENTATION

Group Members	Muhammad Hamza
	Muhammad Huzaifa Attari
	Zeera Khan
	Muhammad Annas

Supervisors	Nadeem Mehmood
	Mukesh Kumar

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1. Abstract

Cerebral aneurysms are localized dilations of intracranial blood vessels whose rupture causes subarachnoid hemorrhage, a condition with high mortality and morbidity. Clinical decision-making around whether an unruptured aneurysm requires intervention or monitoring increasingly draws on **Computational Fluid Dynamics (CFD)**-derived hemodynamic biomarkers — principally Time-Averaged Wall Shear Stress (TAWSS) and the Oscillatory Shear Index (OSI) — alongside demographic and morphological risk scores such as PHASES.

NeuroFlow CFD Analyst is a web-based clinical decision-support dashboard that visualizes these hemodynamic and morphological indicators for patient-specific cerebral aneurysm cases. The system presents a 2D color-mapped hemodynamic heatmap, an interactive 3D vessel/nerve visualization, a live-computed Composite Risk Index with factor-level explainability, supplementary markers (Relative Residence Time and Endothelial Cell Activation Potential), a PHASES clinical risk score, and an exportable clinical report. A DICOM metadata ingestion pipeline (Python, `pydicom`) forms the first real backend component toward a full CFD pipeline. This document describes the project's motivation, design, implementation, and evaluation, and outlines the path toward a fully computed (rather than visualized) hemodynamic simulation backend.

2. Introduction

2.1 Background & Motivation

An intracranial (cerebral) aneurysm is an abnormal bulge in the wall of a brain artery, most commonly forming at arterial bifurcations where hemodynamic stress is highest. Most aneurysms are discovered incidentally during imaging performed for unrelated reasons and never rupture; however, when rupture does occur, it produces a subarachnoid hemorrhage with mortality reported in the range of 30–50%. The central clinical question — and the one this project is built around — is therefore not "does this patient have an aneurysm?" but *"how likely is this specific aneurysm to rupture, and does that risk justify the risk of surgical or endovascular intervention?"*

Traditional risk stratification relied primarily on aneurysm size and patient demographics. Over the past decade, CFD-derived hemodynamic parameters — computed by simulating blood flow through a patient-specific 3D reconstruction of the vessel, typically segmented from MRA (Magnetic Resonance Angiography) or CTA (Computed Tomography Angiography) imaging — have been shown to correlate with rupture status, motivating the development of hemodynamics-aware clinical decision-support tools.

2.2 Problem Statement

Raw CFD output (millions of mesh nodes, each carrying multiple time-varying scalar fields) is not directly usable by a clinician at the point of care. There is a need for a dashboard layer that:

- Translates CFD-derived hemodynamic fields into an intuitive, color-coded spatial visualization;
- Surfaces the specific numeric thresholds that clinical literature associates with elevated rupture risk;
- Combines hemodynamic (flow-based) and morphological/demographic (PHASES-style) risk assessment into a single, explainable composite score, consistent with current multimodal clinical guidance; and
- Produces a clinician-facing report suitable for case review and documentation.

NeuroFlow CFD Analyst was built to address this dashboard layer, using a realistic patient dataset and a clinically-grounded set of derived metrics, with an explicit, documented boundary around which parts of the system are computed live from data versus which represent a scripted visualization of a solver pipeline that a production deployment would replace with a real backend (see Section 9).

3. Objectives

1. Design and implement a patient-centric dashboard for reviewing CFD-derived hemodynamic data across multiple cerebral aneurysm cases.
2. Implement a 2D spatial heatmap visualization mapping Wall Shear Stress-related metrics onto a vessel/aneurysm geometry, toggle-able between TAWSS and OSI display modes.
3. Implement a 3D visualization of the cerebral vessel/nerve network, with risk-driven, localized color mapping rather than a uniform whole-model tint.
4. Derive a live, explainable Composite Risk Index (CRI) from hemodynamic and morphological inputs, rather than using static, pre-assigned risk labels.
5. Extend hemodynamic assessment beyond TAWSS/OSI with supplementary markers used in current CFD rupture-risk literature: Relative Residence Time (RRT) and Endothelial Cell Activation Potential (ECAP).
6. Implement a validated, published demographic/morphological risk score (PHASES) as an independent, complementary risk signal alongside the hemodynamic CRI.
7. Build a DICOM metadata ingestion component using real medical-imaging parsing (`pydicom`) as the first backend step toward a full CFD pipeline.
8. Generate a structured, exportable clinical report summarizing all computed metrics against their clinical thresholds.

4. Literature Review

4.1 CFD in Cerebral Aneurysm Research

Patient-specific hemodynamic CFD analysis of cerebral aneurysms follows a well-established pipeline: (1) medical image acquisition (MRA/CTA), (2) vessel segmentation to produce a 3D surface model, (3) surface and volumetric mesh generation, (4) assignment of patient-specific or population-averaged boundary conditions (inlet velocity waveform, outlet pressure, blood density/viscosity), (5) numerical solution of the unsteady Navier–Stokes equations over several cardiac cycles, and (6) post-processing to extract wall-based hemodynamic fields. **SimVascular** is the most widely cited fully open-source implementation of this entire pipeline, and is frequently paired with the **Vascular Modeling Toolkit (VMTK)** for vessel-lumen extraction and centreline-based geometric analysis; solvers such as **OpenFOAM** are also used for the Navier–Stokes solve stage. NeuroFlow CFD Analyst's simulated six-stage pipeline (Section 6.7) mirrors this exact stage sequence for demonstration purposes.

4.2 Hemodynamic Wall Parameters

Four wall-based scalar fields are used in this project, all derived from the same underlying quantity — the instantaneous Wall Shear Stress (WSS) vector field on the vessel surface, integrated or statistically summarized over one cardiac cycle:

Parameter	Definition	Rupture Association
TAWSS Time-Averaged Wall Shear Stress	Magnitude of the WSS vector averaged over one full cardiac cycle.	Low TAWSS is associated with flow stagnation and endothelial dysfunction at the aneurysm wall.
OSI Oscillatory Shear Index	Measures the degree to which the WSS vector direction reverses during the cycle; ranges 0–0.5.	High OSI indicates disturbed, non-unidirectional flow linked to wall degeneration.
RRT Relative Residence Time	Combines TAWSS and OSI: $RRT \propto \frac{1}{(1-2 \cdot OSI) \cdot TAWSS}$	Approximates how long blood constituents reside near the wall; elevated RRT is associated with atherogenic, pro-thrombotic conditions.
ECAP Endothelial Cell Activation Potential	$ECAP = OSI / TAWSS$	Values above ≈ 1 indicate the oscillatory component dominates over mean shear, a pattern associated with elevated rupture risk in CFD-based rupture-risk studies.

Combining more than one of these markers is explicitly recommended in current literature; single-metric (TAWSS- or OSI-only) assessment is noted to be an oversimplification of the true multi-factorial hemodynamic environment at the aneurysm wall.

4.3 Clinical Risk Scoring (PHASES)

The **PHASES** score (Greving et al., 2014) is a validated, points-based score for estimating the 5-year cumulative rupture risk of an unruptured intracranial aneurysm from six independent factors: **P**opulation/ethnicity, **H**ypertension, **A**ge, aneurysm **S**ize, **E**arlier subarachnoid hemorrhage from a different aneurysm, and **S**ite of the aneurysm. It is a purely demographic/morphological score — it does not use any hemodynamic (flow) information. Current clinical guidance recommends combining scores like PHASES with imaging-derived and hemodynamic assessment for a fuller risk picture, since morphological/demographic scores alone have been shown in follow-up studies to under-perform when used as the sole predictor of subsequent rupture. This is the rationale for reporting the PHASES score in this project as a distinct, complementary panel alongside — not merged into — the CFD-derived Composite Risk Index.

4.4 Related Open-Source Tools

- **SimVascular** — the only fully open-source, end-to-end pipeline from medical image segmentation through patient-specific CFD simulation.
- **VMTK (Vascular Modeling Toolkit)** — vessel-lumen segmentation and centreline extraction, commonly used upstream of meshing.
- **OpenFOAM** — open-source finite-volume CFD solver, applicable to the Navier–Stokes solve stage of the pipeline.
- **Cornerstone.js** — open-source JavaScript library (used to build the OHIF radiology viewer) for standards-compliant, web-based DICOM image rendering.

5. System Design & Architecture

5.1 Technology Stack

Layer	Technology	Purpose
Structure & Styling	HTML5, CSS3	Dashboard layout, responsive panels, print-styled clinical report
Application Logic	Vanilla JavaScript (ES6+)	State management, DOM rendering, metric computation, pipeline orchestration
2D Visualization	HTML5 Canvas API	Hemodynamic heatmap, mesh/convergence animation panels
3D Visualization	Three.js (WebGL), GLTFLoader, OrbitControls	Interactive 3D nerve/vessel model with per-vertex risk coloring
Backend Prototype	Python 3, <code>pydicom</code>	Real DICOM metadata extraction (Patient ID, Modality, Slice Thickness, etc.)
Icons / Typography	Font Awesome, Google Fonts (Ubuntu)	UI iconography and typography

5.2 Architectural Overview

The application follows a client-side, single-page architecture with a clear separation between the **data model** (an in-memory patient database keyed by patient ID, each record holding raw hemodynamic zone data, morphology, and demographics), **derived-metric functions** (pure functions that compute CRI, RRT, ECAP, and PHASES from the raw data on every render, so displayed values can never drift out of sync with their inputs), and the **rendering layer** (Canvas 2D heatmap, Three.js 3D viewer, and DOM-based gauge/report components), all coordinated by a central patient-selection state.

5.3 Data Flow

1. User selects a patient card, or uploads a DICOM/MRA file (triggering the simulated ingestion pipeline).
2. The active patient record (hemodynamic zones, morphology, demographics) is set as global state.
3. Derived-metric functions recompute CRI, RRT, ECAP, and PHASES from that record.
4. The 2D canvas, 3D viewer, telemetry gauges, and patient list all re-render from the same recomputed values.

5. On "Expand Case Review," the same values are formatted into the clinical report view and made exportable via the browser print pipeline.

6. Implementation

6.1 Patient Dashboard & Data Model

Each patient record stores an array of anatomical **zones** (Parent Artery Inlet, Parent Artery Outlet, Aneurysm Neck, Aneurysm Dome), each carrying its own TAWSS and OSI reading — mirroring how a real CFD post-processor reports results at discrete mesh nodes rather than as a single aggregate value. Patient cards in the sidebar display the live-computed Composite Risk Index and aneurysm diameter, and include a delete control (with confirmation, and safeguards against removing the last remaining case) so uploaded or mock cases can be managed without a page reload.

6.2 2D Hemodynamic Heatmap

The heatmap is drawn on an HTML5 Canvas as a parent-artery tube with an aneurysm neck/dome rising from its midpoint. Each zone's color is computed by linearly interpolating between a stable-blue and critical-red hex color, using a mode-dependent normalization: TAWSS values are inverted (low TAWSS → red) and normalized against a 0.15–1.5 Pa range; OSI values are normalized directly against a 0.03–0.35 range. Healthy parent-artery zones have their color factor dampened ($\times 0.2$) so that non-aneurysm anatomy remains visually calm even under the same color scale. Hovering any node opens a tooltip with its exact TAWSS/OSI readings and a rupture-vulnerability flag where applicable.

6.3 3D Neuro-Vascular Visualization

A supplied `.glb` 3D asset was found on inspection to contain three sub-meshes: a high-density ($\approx 450,000$ -vertex) branching vessel/nerve network, and two lower-relevance solid brain-shape meshes bundled in the same file. Only the vessel/nerve mesh is rendered; the brain-shape meshes are hidden, since they would otherwise visually occlude the vessel network entirely. Risk coloring is applied **per vertex** rather than as a uniform tint: a "hotspot" region is approximated as the topmost cluster of vertices (consistent with where the aneurysm dome is positioned in the 2D layout), and each vertex's redness is scaled by both its proximity to that hotspot and the patient's live TAWSS/OSI-derived risk factor — so the vessel network stays blue overall, with only the clinically critical region shifting toward red. The viewer supports mouse-driven orbit/zoom via Three.js `OrbitControls`.

6.4 Composite Risk Index & Explainability

The Composite Risk Index (CRI) is computed live, not stored, as a weighted sum of four normalized sub-scores:

```

CRI = 0.35·TAWSS_score + 0.30·OSI_score + 0.20·Diameter_score +
0.15·AspectRatio_score

TAWSS_score    = clamp01((1.5 - TAWSS) / (1.5 - 0.15)) × 100
OSI_score      = clamp01((OSI - 0.03) / (0.35 - 0.03)) × 100
Diameter_score = clamp01((Diameter - 2.0) / (10.0 - 2.0)) × 100
Aspect_score   = clamp01((AspectRatio - 0.7) / (2.5 - 0.7)) × 100

```

Scores of 75 or above are classified High Rupture Risk, 45–74 Moderate, and below 45 Stable/Low Risk. Because every consumer of this score (patient cards, the composite gauge, the report modal) calls the same function against the same underlying zone data, the displayed number can never disagree with the raw metrics shown elsewhere in the dashboard. A breakdown panel beneath the gauge renders each factor's percentage contribution as a labeled bar, directly answering "how was this number produced?" for the benefit of clinical (and academic) review.

6.5 Advanced Hemodynamic Markers

RRT and ECAP (defined in Section 4.2) are computed from the same dome TAWSS/OSI values and displayed as two additional radial gauges, with alert thresholds of $RRT > 3.0 \text{ Pa}^{-1}$ and $ECAP > 1.0$ respectively. The RRT formula's denominator is numerically guarded against the $OSI \rightarrow 0.5$ singularity.

6.6 PHASES Clinical Risk Score

Each patient record carries a `demographics` object (age, hypertension status, prior SAH history, population category, aneurysm site). A dedicated scoring function reproduces the published PHASES point table and 5-year cumulative risk-percentage lookup, and renders both the point breakdown and resulting percentage in a dedicated card, explicitly labeled as a demographic/morphological score used *alongside* the hemodynamic CRI rather than folded into it — consistent with the multimodal approach described in Section 4.3.

6.7 Simulated CFD Pipeline (6-Stage)

Uploading a DICOM/MRA file triggers a six-stage, timed visualization mirroring the real-world pipeline described in Section 4.1: (1) DICOM ingestion & metadata parsing, (2) AI segmentation, (3) surface/volume mesh generation, (4) boundary-condition assignment, (5) CFD solver execution with a live residual-convergence plot, and (6) metric extraction & report compilation. Each stage renders its own Canvas-based animation (an axial-scan overlay, a rotating wireframe mesh, a pulsatile inflow waveform, and a logarithmic convergence curve respectively) alongside a scrolling terminal log. On completion, a new patient record is generated with hemodynamic

values seeded from the filename pattern, and the same live-computed CRI/RRT/ECAP/PHASES pipeline is applied to it exactly as it is for the seed patients.

6.8 Clinical Report Generation

The "Expand Case Review" report compiles all computed metrics (TAWSS, OSI, RRT, ECAP, diameter, aspect ratio, Composite Risk Score, and the full PHASES breakdown) into a structured table against their clinical thresholds, alongside a snapshot of the current heatmap and a narrative clinical assessment. The report is exportable via the browser's print pipeline to PDF.

6.9 DICOM Metadata Parsing (Python Backend)

A standalone Python module (`neuroflow-local-pipeline/`) uses `pydicom` to safely extract standard DICOM header fields (`PatientID`, `Modality`, `StudyDate`, `Rows`, `Columns`, `SliceThickness`) via defensive `getattr()` access, with physiological constants (blood density 1060 kg/m³, viscosity 0.0035 Pa·s) centralized in a configuration module. This is the one component of the pipeline that performs genuine, non-simulated processing on real DICOM input, and is structured so it can be dropped directly into an asynchronous task worker (e.g., Celery) in a future backend-integrated version.

7. Results & Screenshots

The following figures document the working application across its major views, captured from a live session of the dashboard.

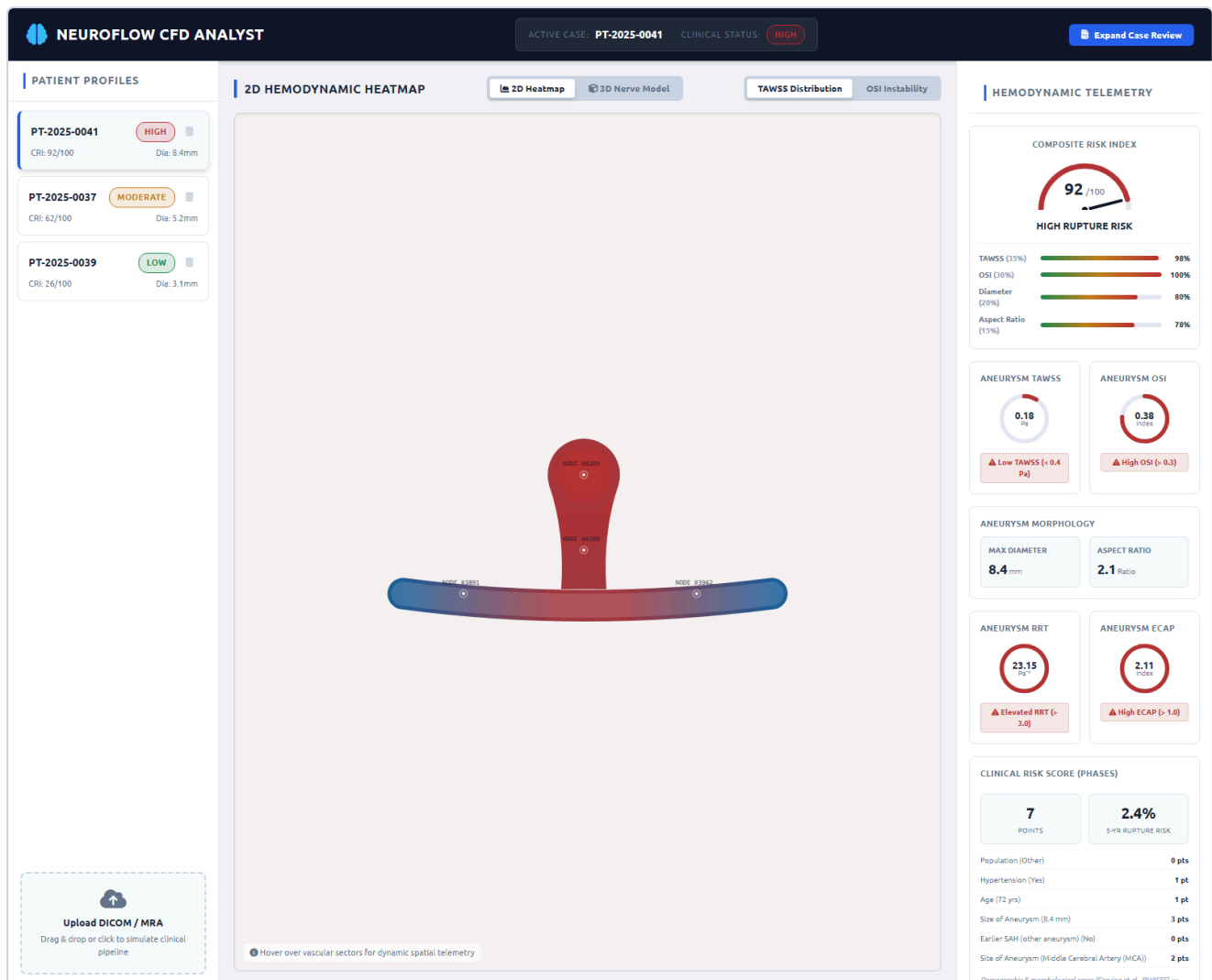


Figure 7.1 — Main dashboard: 2D Hemodynamic Heatmap for a high-risk patient (PT-2025-0041, CRI 92/100). The aneurysm dome and neck render deep red under low TAWSS, while the parent artery remains blue. The right-hand telemetry panel shows the Composite Risk Index with its factor-level breakdown, TAWSS/OSI/RRT/ECAP gauges, morphology, and the PHASES clinical risk score.

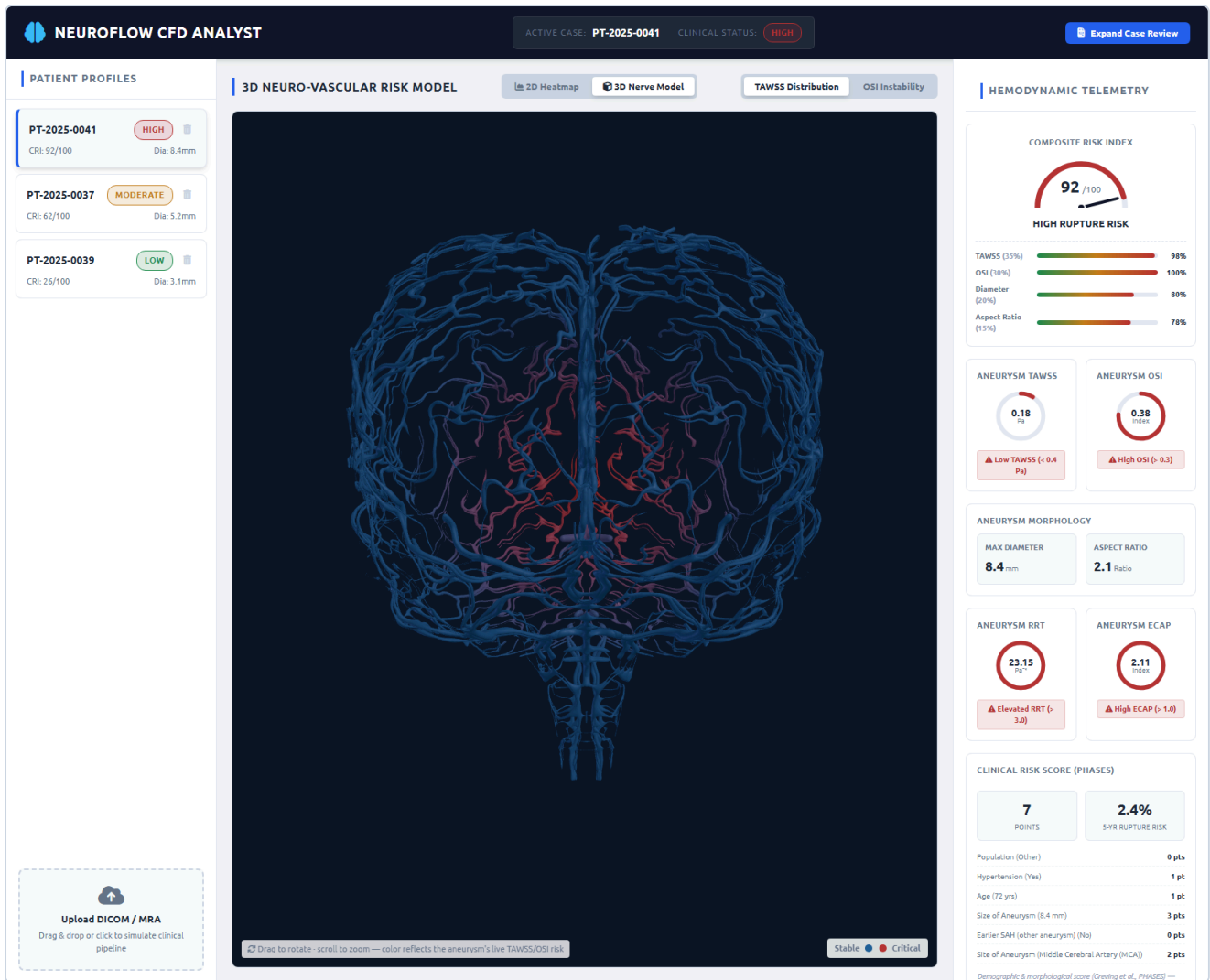


Figure 7.2 — 3D Neuro-Vascular Risk Model view for the same high-risk patient. Per-vertex coloring keeps the vessel/nerve network blue overall, with only the region near the approximated aneurysm site shifting to red — scaled by the patient's live TAWSS/OSI risk factor.

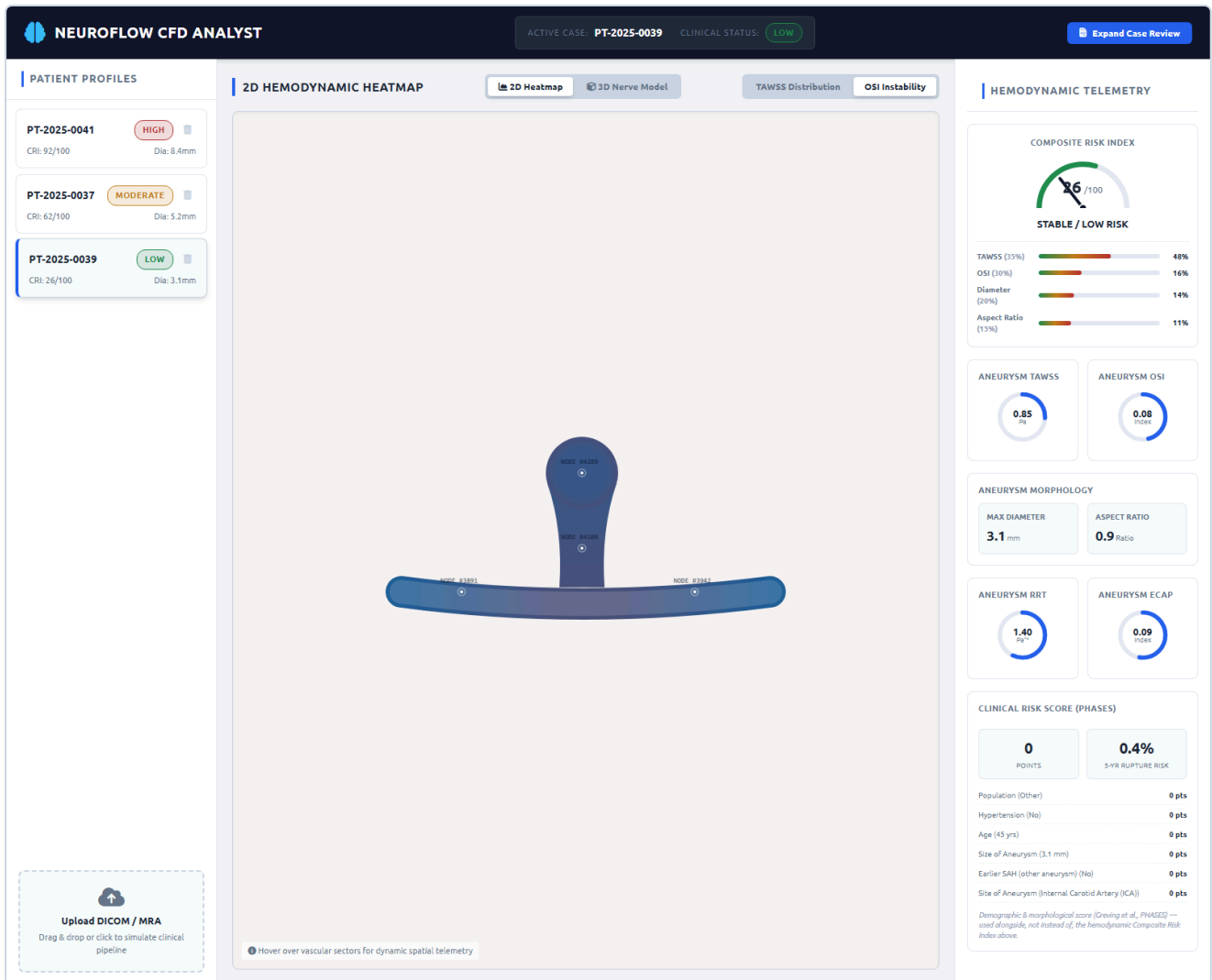


Figure 7.3 — 2D heatmap for a low-risk patient (PT-2025-0039, CRI 26/100) in OSI Instability display mode, showing the map-mode toggle and the corresponding stable (blue-dominant) coloring.

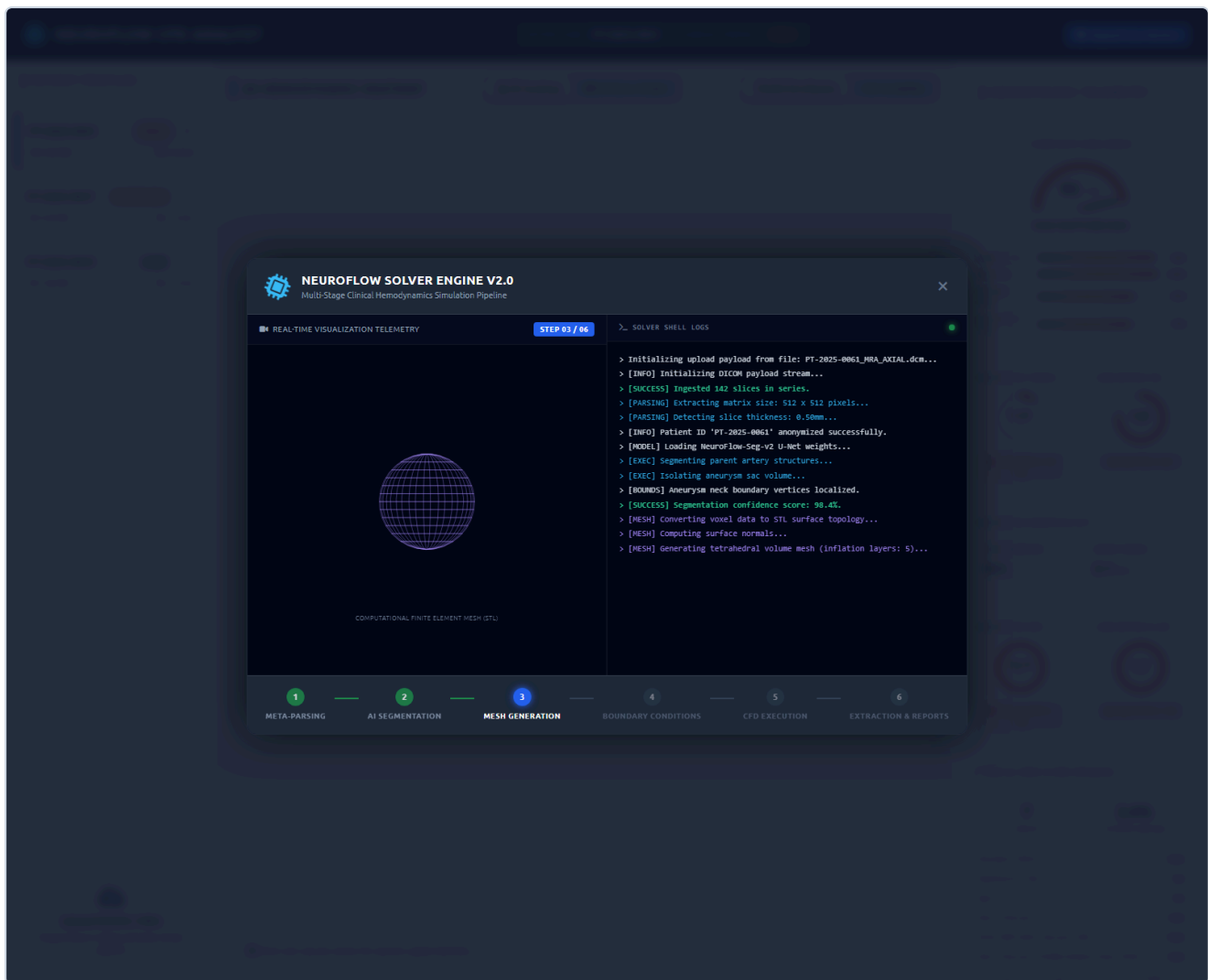


Figure 7.4 — Simulated CFD pipeline, Stage 3 (Mesh Generation), showing the rotating wireframe animation and scrolling solver shell log alongside the six-stage progress tracker.

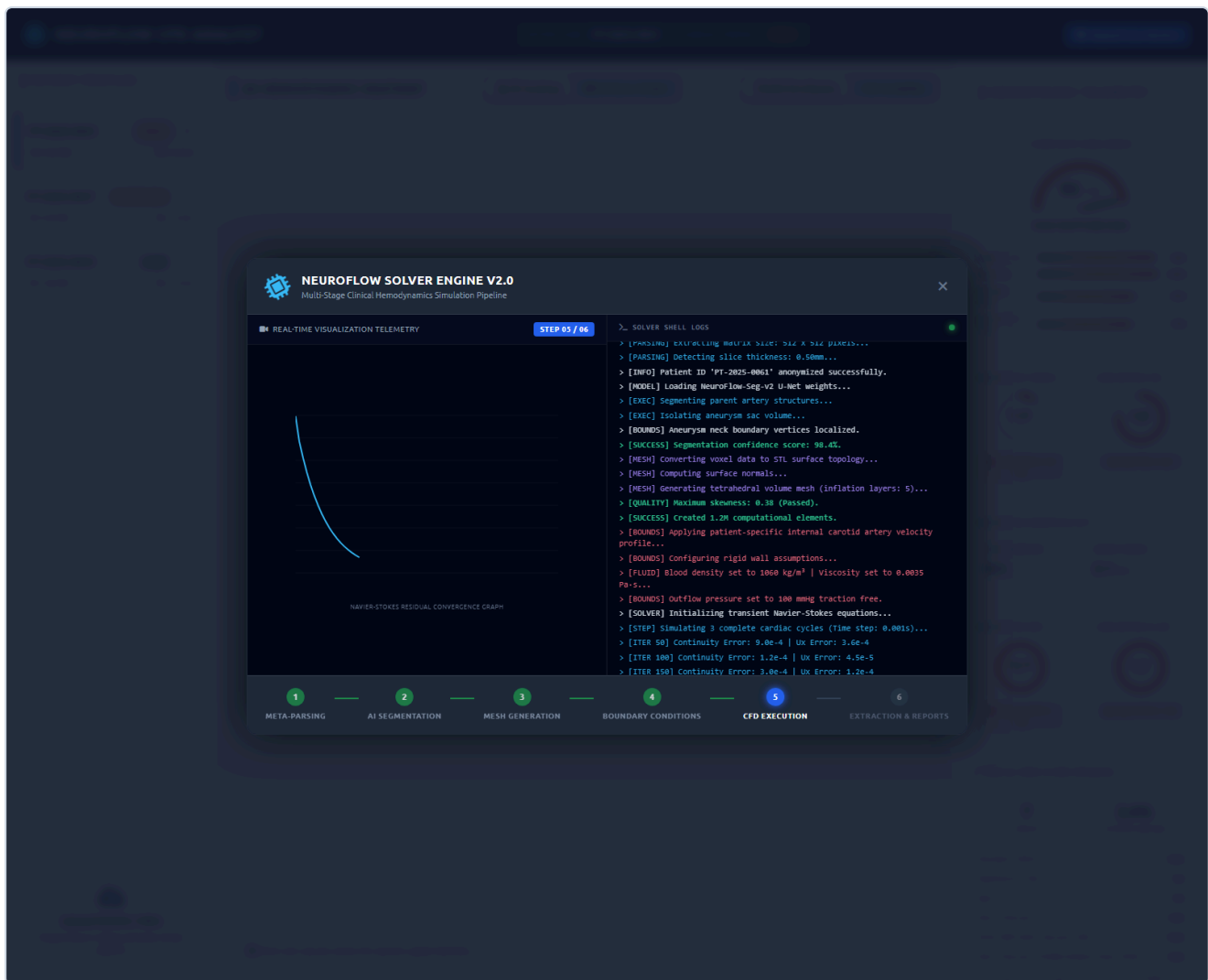


Figure 7.5 — Simulated CFD pipeline, Stage 5 (CFD Execution), showing the live Navier–Stokes residual convergence graph and iteration log.

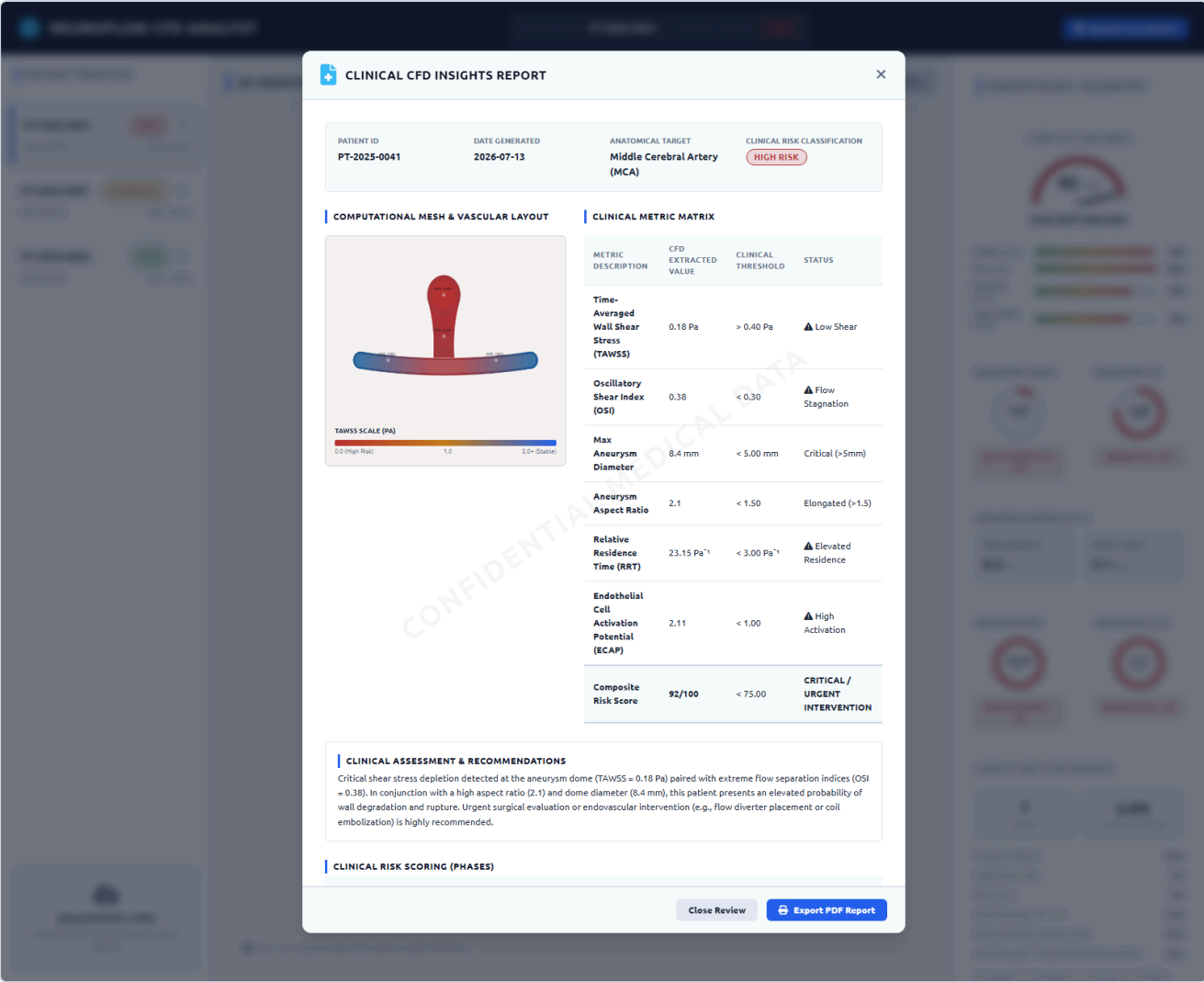


Figure 7.6 — Exportable Clinical CFD Insights Report for the high-risk patient, showing the full Clinical Metric Matrix (including RRT and ECAP) against their clinical thresholds, and the dynamic anatomical target derived from the patient's PHASES site data.

8. Testing & Validation

The application was validated through automated, browser-driven end-to-end testing (Playwright with headless Chromium) covering:

- **Cross-view consistency** — verifying that switching patients or toggling TAWSS/OSI mode updates the 2D heatmap, 3D model, all gauges, and the report modal in agreement, with no stale or conflicting values.
- **Numerical correctness** — RRT, ECAP, Composite Risk Index, and PHASES score outputs were independently hand-calculated for each seed patient and cross-checked against the rendered UI values.
- **Layout regression testing** — automated bounding-box measurements were used to verify that UI elements (e.g., the CFD convergence graph, the extraction-stage checklist) render correctly centered within their containers.
- **Interaction flows** — the full upload → six-stage pipeline → new patient record flow, patient deletion (including edge cases such as deleting the active patient or the last remaining case), and report export were exercised end-to-end.
- **Console/runtime error monitoring** — every test run captured browser console and page errors; all reported results in this document reflect a zero-error state.

9. Limitations

Transparency note. This section is included deliberately: an accurate account of what is simulated versus computed is, in the authors' view, part of correct academic documentation for this project.

- **The segmentation, meshing, and Navier–Stokes solver stages are simulated.** The six-stage upload pipeline (Section 6.7) is a scripted, timed visualization that reproduces the real-world CFD workflow's stages, log format, and terminology, but does not perform actual image segmentation, mesh generation, or fluid-dynamics computation.
- **Patient hemodynamic data is a curated demonstration dataset.** The three seed patients represent illustrative High/Moderate/Low risk profiles; values for newly "uploaded" cases are seeded from a filename pattern rather than derived from the uploaded file's actual image content.
- **The 3D hotspot region is a geometric approximation.** The supplied 3D asset carries no per-vertex clinical zone labels, so the "critical region" highlighted in the 3D view is inferred from vertex position (Section 6.3) rather than verified anatomical correspondence.
- **No persistence layer.** All patient data is held in-memory and is reset on page reload; there is no database or server-side storage.

What is real and independently verifiable in the current system: the Composite Risk Index, RRT, ECAP, and PHASES score computations (Sections 6.4–6.6); the DICOM metadata parser (Section 6.9); and the full rendering/visualization pipeline.

10. Future Work

1. **Real segmentation and meshing backend** integrating an open-source pipeline such as SimVascular with VMTK for vessel extraction (Section 4.4), replacing the simulated Stage 2–3 animations with genuine 3D reconstruction from uploaded imaging.
2. **Real CFD solver integration** via OpenFOAM, executing the actual unsteady Navier–Stokes solve against patient-specific boundary conditions rather than a scripted convergence curve.
3. **A trained ML surrogate model** (e.g., a lightweight regression or graph neural network) predicting hemodynamic outcomes directly from morphology, offering near-real-time estimates without a full CFD solve for triage use cases — mirroring current research directions in CFD surrogate modeling.
4. **A real DICOM image viewer** using Cornerstone.js, replacing the current metadata-only text parsing with standards-compliant medical image rendering.
5. **Validation against public benchmark datasets** such as AneuriskWeb or the newer Aneumo dataset, to quantitatively assess the Composite Risk Index's agreement with documented rupture outcomes.
6. **Mesh-convergence and uncertainty quantification studies**, standard practice in published patient-specific CFD work, once a real solver is integrated.
7. **Longitudinal, multi-visit patient tracking**, enabling the dashboard to show hemodynamic and morphological change over time for the same patient — consistent with the cardiovascular "digital twin" direction in current research.

11. Conclusion

NeuroFlow CFD Analyst demonstrates a complete, clinically-informed dashboard layer for cerebral aneurysm hemodynamic risk assessment: a live-computed, explainable Composite Risk Index; supplementary CFD-literature markers (RRT, ECAP); an independent, validated demographic risk score (PHASES); 2D and 3D visualizations driven by the same underlying data; and an exportable clinical report. The project's real backend component — DICOM metadata parsing — establishes a concrete foundation for the segmentation, meshing, and solver integration work identified in Section 10 as the path to a fully computed (rather than visualized) hemodynamic simulation system. Documenting clearly which components are computed versus simulated (Section 9) was treated as a first-class design requirement throughout the project, consistent with the standards expected of clinically-oriented software.

12. References

1. Greving, J. P., et al. (2014). *Development of the PHASES score for prediction of risk of rupture of intracranial aneurysms: a pooled analysis of six prospective cohort studies*. The Lancet Neurology.
2. Meng, H., Tutino, V. M., Xiang, J., & Siddiqui, A. (2014). *High WSS or Low WSS? Complex interactions of hemodynamics with intracranial aneurysm initiation, growth, and rupture: toward a unifying hypothesis*. American Journal of Neuroradiology.
3. CFD-based hemodynamic markers for cerebral aneurysm rupture-risk prediction. arXiv:2504.10524.
4. Relative Residence Time and Endothelial Cell Activation Potential in cerebral aneurysm hemodynamics. PMC11907627.
5. Updahyaya et al. Semi-automated open-source pipeline for anterior brain vasculature CFD simulation. Scientific Reports (2024).
6. Updegrove, A., et al. *SimVascular: An Open Source Pipeline for Cardiovascular Simulation*. Annals of Biomedical Engineering.
7. Cornerstone.js — Open Health Imaging Foundation. cornerstonejs.org
8. Aneumo: a large-scale multimodal aneurysm CFD and deep-learning benchmark dataset. arXiv:2505.14717.
9. AneuriskWeb project repository — Emory University Center for Computational and Modeling of Engineering.
10. Evaluating CFD Uncertainty via Grid Convergence Index in patient-specific aorta models. arXiv:2309.05509.